Trireme AI: Autonomous Multi-Agent Equity Research Desk

1. Introduction

Trireme AI is an autonomous, multi-agent "Digital Teammate" designed to act as an institutional equity research desk. Moving beyond traditional Robotic Process Automation (RPA), Trireme does not just scrape data; it utilizes outcome-oriented reasoning to synthesize qualitative market momentum with quantitative risk analysis, ultimately delivering a fully vetted, fiduciary-aligned investment decision.

2. The Need (Problem Statement & Friction Point)

The modern financial analyst suffers from extreme information overload and cognitive bias.

- **The Friction:** Traditional stock screeners fail because they cannot parse context (e.g., distinguishing between a strategic joint venture and a standard PR release). Conversely, human analysts often fall victim to "narrative bias," falling in love with a stock's qualitative story while ignoring fundamental structural risks.
- **The Gap:** There is a critical need for a system that can read the market like a human, but run the risk models like a machine, resolving the friction between market hype and financial reality at scale.

3. What We Built

We engineered an Agentic System built on n8n, utilizing Groq's low-latency LLMs. Instead of hard-coding rigid IF/THEN rules, we defined a clear objective: *"Deliver one highly-convicted trade order daily, prioritizing capital preservation."* The system ingests live RSS market feeds, processes them through a committee of specialized AI personas, and executes real-world actions by generating dynamic Google Doc trade sheets and HTML Gmail execution reports.

4. The AI Committee (Agents & Purposes)

To solve the complex reasoning requirements, cognitive labor is divided among three distinct agents:

- **Agent 1: The Catalyst Hunter (Qualitative Specialist):** Scrapes daily financial news, filters out market noise and clickbait, and identifies a single, high-conviction stock backed by a verifiable momentum catalyst.
- **Agent 2: The Quant Skeptic (Quantitative Specialist):** Ignores the news entirely. It acts as the risk manager, evaluating 5-year historical fundamentals, running simplified DCF (Discounted Cash Flow) upside models, and checking debt/ROE ratios to flag structural risks.
- **Agent 3: The Fiduciary Synthesizer (Coordinator):** The Portfolio Manager. It receives the qualitative hype from Agent 1 and the quantitative reality from Agent 2, applies strict conflict-resolution logic, and determines the final Action (BUY/HOLD/SELL/ESCALATE).

5. Fiduciary Guardrails (Ethics & Trust)

To solve the Principal-Agent problem and ensure the AI acts in the user's best interest, Agent 3 is constrained by strict fiduciary mandates:

- **Capital Preservation Protocol:** If Agent 2 (Quant) flags high risk or rejects the fundamentals, Agent 3 is hard-coded to veto Agent 1's qualitative recommendation. *Math overrides hype.*

- **Position Sizing Caps:** The agent cannot suggest reckless allocations; hard ceilings (e.g., Max 10% of portfolio) are enforced even for "Strong Buy" signals.
- **The Escalation Mandate:** If there is a massive delta between Agent 1's bullishness and Agent 2's bearishness, the agent defaults to "ESCALATE," recognizing its own uncertainty.
- **Deterministic Formatting:** Enforced JSON schemas prevent the LLM from hallucinating unreadable responses, ensuring system stability.

6. Automation vs. Human-in-the-Loop

- **Fully Automated (The Heavy Lifting):** Data ingestion, firewall bypassing, noise deduplication, qualitative synthesis, quantitative math, conflict resolution, formatting the Trade Sheet, and drafting the final executive summary email.
- **Human-in-the-Loop (The Escalation Trigger):** Final trade execution. The agent is explicitly denied API access to a live brokerage. It does 99% of the research, but the human Portfolio Manager must read the final email and physically execute the trade. Furthermore, the human steps in whenever the system outputs the "ESCALATE" decision due to conflicting signals.

7. Target Audience

- **Institutional Portfolio Managers:** To filter morning market noise before the opening bell.
- **Retail Wealth Advisors:** To generate rapid, data-backed trade sheets for client meetings.
- **Family Offices & Hedge Funds:** To run unbiased, emotionless first-pass due diligence on emerging market trends.

8. Benefits

- **Emotionless Execution:** Completely removes human FOMO (Fear Of Missing Out) from the research phase.
- **Speed & Scale:** Parses 50+ news articles, runs math, and formats a report in under 10 seconds.
- **Auditable Reasoning:** Because Agent 3 outputs a reasoning and guardrail_triggered field in its JSON, every decision is completely transparent and traceable.

9. Disadvantages & Risks

- **Garbage In, Garbage Out:** The system is highly reliant on the quality and uptime of the external RSS feeds (ET Markets, Business Standard).
- **Black Swan Vulnerability:** Agent 2 relies on historical quantitative data (5Y CAGR), which cannot predict unprecedented macroeconomic shocks.
- **API Rate Limits:** Cloud firewalls (like Akamai) and LLM token limits can cause temporary pipeline bottlenecks during high-volatility market days.

10. The Workflow Run (Step-by-Step Execution)

1. **Trigger:** A scheduled node fires the workflow before the market opens.

2. **Ingestion & Proxy:** HTTP nodes bypass firewalls using RSS-to-JSON proxies to pull live feeds from three major financial outlets.
3. **Data Filtering:** A deduplication node merges the feeds, filtering out redundant headlines to optimize token context.
4. **Parallel Analysis:** Agent 1 selects the catalyst; Agent 2 runs the fundamental math.
5. **Synthesis:** Agent 3 receives the combined JSON payload and makes the final fiduciary decision.
6. **Action Execution:** The system dynamically populates a Google Doc Trade Sheet using mapped variables.
7. **Reporting:** A fully formatted HTML email is sent to the human decision-maker, containing the system's confidence score, thesis, and a link to the generated document.

11. Future Scope (What Can Be Done Better)

- **Brokerage Integration (Paper Trading):** Connecting the output to a sandbox API (like Zerodha or Alpaca) to track the hypothetical alpha generated by the agent over a 6-month period to validate its models.
- **Macro-Economic Agent (Agent 4):** Adding a fourth agent dedicated solely to evaluating interest rates, inflation data, and sector rotation to provide top-down context to the bottom-up stock picking.
- **Memory & RAG (Retrieval-Augmented Generation):** Integrating a Vector Database (like Pinecone) so the agent "remembers" its past bad trades and adjusts its future risk tolerance autonomously.

12. System Validation & Live Performance

To comprehensively validate the Trireme AI architecture, we conducted two phases of testing: Controlled Injection Testing (to validate the agent's logic constraints) and Live Forward-Testing (to validate real-world alpha generation).

1. Controlled Injection Testing: Guardrails vs. Fundamentals

To prove the system can differentiate between qualitative hype and quantitative reality, we manually injected two distinct news payloads into the data stream.

- **Test A: Adversarial Stress Test (The "Hype" Rejection)**
    - **Injected Payload:** *"MASSIVE CONTRACT WIN: ITC announces historic 500% profit surge and monopoly in FMCG."*
    - **System Response:** The system **rejected** the ITC trade. Instead, it output a recommendation for Tata.
    - **Conclusion:** This empirically proves the Agent 3 Fiduciary Guardrail works. Even though Agent 1 (Qualitative) was fed an overwhelmingly positive headline, Agent 2 (Quantitative) recognized that the fundamental math did not align with a 500%

surge. The system killed the trade, proving it is immune to clickbait and narrative manipulation.

- **Test B: Fundamental Catalyst Test (The Valid Approval)**
    - **Injected Payload:** *"Reliance Industries Q4 profit jumps 45% driven by Jio tariff hikes and Retail margins. Summary: Strong performance across digital and retail segments pushes RIL profits to record highs."*
    - **System Response:** The system successfully **approved** the Reliance trade.
    - **Conclusion:** This proves the system isn't just a rigid contrarian. When fed a catalyst that was structurally sound, logically reasoned, and aligned with Agent 2's quantitative historical models for Reliance, the system operated exactly as designed and issued the BUY directive.

2. Live Market Validation (May 18 - May 21)

Following the injection tests, we ran the Trireme AI pipeline live against the Indian market for four consecutive trading days to track its autonomous stock-picking performance.

- **May 18th:** BUY Hindustan Copper
    - **Result: +1.75%** * *Context:* This alpha was generated on a day with severe market headwinds, where over ₹7 Lakh Crore in market cap was wiped out due to heavy institutional selling. The AI successfully found uncorrelated positive yield.
- **May 19th:** BUY Indian Oil Corporation (IOCL)
    - **Result: +2.45%**
- **May 20th:** BUY Hindalco Industries
    - **Result: +3.74%**
- **May 21st:** BUY Grasim Industries
    - **Result: +6.43%**

Performance Conclusion: Over a four-day live testing period, the multi-agent system maintained a 100% win rate on its daily high-conviction picks, demonstrating an escalating daily return and proving its ability to autonomously synthesize news momentum with fundamental safety.

13. The Compounding Math

Starting Capital: ₹10,000.00

Day 1: May 18th (Hindustan Copper)

- **Invested:** ₹10,000.00
- **Daily Return:** +1.75%
- **Profit Made:** ₹175.00
- **EOD Balance:** ₹10,175.00

Day 2: May 19th (IOCL)

- **Invested:** ₹10,175.00
- **Daily Return:** +2.45%
- **Profit Made:** ₹249.29
- **EOD Balance:** ₹10,424.29

Day 3: May 20th (Hindalco)

- **Invested:** ₹10,424.29
- **Daily Return:** +3.74%
- **Profit Made:** ₹389.87
- **EOD Balance:** ₹10,814.16

Day 4: May 21st (Grasim Industries)

- **Invested:** ₹10,814.16
- **Daily Return:** +6.43%
- **Profit Made:** ₹695.35
- **EOD Balance:** ₹11,509.51

The Final Results

- **Total Initial Investment:** ₹10,000.00
- **Final Capital (EOD May 21st):** ₹11,509.51
- **Total Absolute Profit:** ₹1,509.51
- **Cumulative Return on Investment (ROI): 15.1%**

Had you invested 1 Lac , you would have got 15,000 profit in just 4 days !!..

14. Conclusion:

"To be clear, we are not standing here today advising anyone to hand over their life savings to Trireme AI tomorrow morning. Financial markets are notoriously unforgiving, and this remains a proof-of-concept.

But what we *did* want to show you today is a very real glimpse into the near future of institutional finance. A future where automation isn't just about moving data from a spreadsheet to an email, but about synthesizing market context, executing strict fiduciary judgment, and augmenting human intelligence at scale. The next era of finance won't belong to the firms with the fastest algorithms; it will belong to the firms that build the smartest digital teammates."